

Critical Flaw in Global NCAP

“The Yawning Chihuahua”

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1 Introduction

Global NCAP added a little bit of well-intentioned *jugaad* to their 2022 protocols to punish cheating, but a silly flaw in the criteria penalises even cars that should not be.

The adult star rating depends on (among other things) the total of the frontal offset and side mobile barrier crash test scores. But what happens when a car does exceptionally well in one test and not so well in the other? No one wants cars to get a good star rating despite having an underwhelming result in one of the tests.

That’s why GNCAP says that if the front and side impact test scores differ too much, they will take away 1 star from the final rating.

The big trouble? This penalises cars for having a misleading score even if the star rating has already been decided by some other factor like pedestrian protection or curtain airbag fitment – which are not related to score.

The recently rated Maruti Suzuki Ertiga doesn’t have curtain airbags, which means it simply can’t get more than 2 stars no matter how well it does in crash tests. The Ertiga’s score is in the 4-star band, but one can see how this could be misleading, as the score is majorly just from side impact and not frontal impact. So even if it meets all feature requirements, the Ertiga will only be allowed max. 3 stars, and it’s easy to understand why.

What *is* deceiving is to reduce its star rating from 2 stars to 1 just because its total score has unfairly crossed into the 4-star band, when **that score isn’t reflecting in the star rating at all.**

Even worse, this can actually penalise cars for improving. Renault are lucky the Triber performed as badly as it did in the side impact test. It could have wound up with 1 star as well, had its side impact chest protection been better.

2 Proof

Here I demonstrate a small but commonly occurring sub-case of the problem where the vehicle model does not meet additional requirements but also does not have any ‘poor’ly protected body regions. A weaker claim is made and proved here, but extending it to other cases is trivial.

Let $\mathbb{V}$ be the Euclidean 2-space and $rating : \mathbb{V} \rightarrow \mathbb{Z}^+$ be the function mapping the vector $(front, side)$ of front and side impact scores to their Global NCAP adult star rating.

Claim:

$$\exists (f_1, s_1), (f_2, s_2) \in \mathbb{V} : f_1 = f_2, s_1 < s_2, rating(f_1, s_1) > rating(f_2, s_2) \quad (1)$$

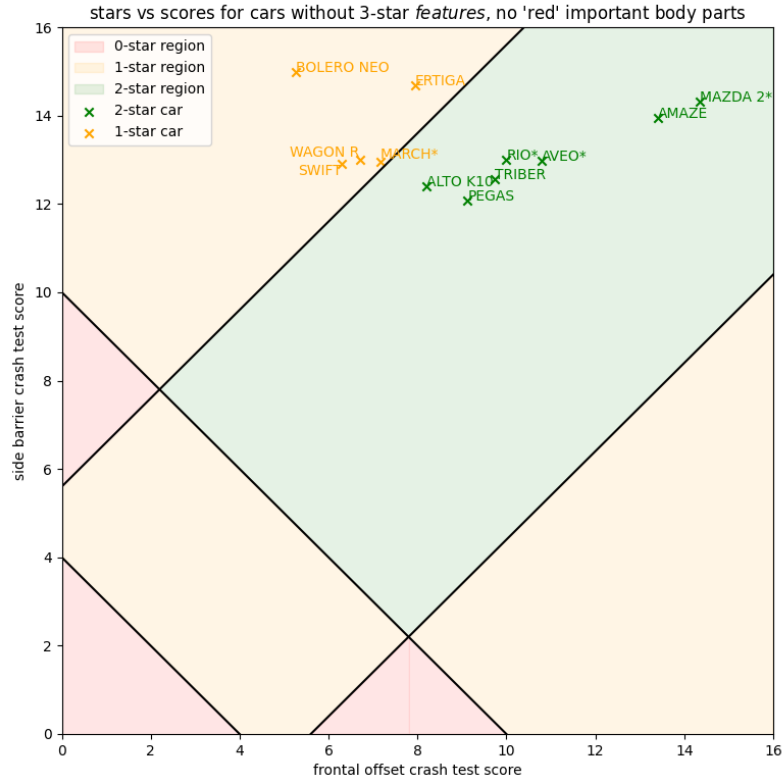
Note the strict inequality and the fact that $rating$ maps to integers. This is equivalent to the claim that under some conditions, when two cars are identical barring their side impact protection, the better performing car could be awarded fewer stars.

The closed form definition for “ $rating$ ” is:

$$rating(x, y) = \begin{cases} 2 & \text{if } x + y \geq 10 \text{ and } |x - y| < 5.6 \\ 1 & \text{if } 4 \leq x + y < 10 \text{ and } |x - y| < 5.6 \\ 0 & \text{if } x + y < 4 \text{ and } |x - y| < 5.6 \\ 1 & \text{if } x + y \geq 10 \text{ and } |x - y| \geq 5.6 \\ 0 & \text{if } x + y < 10 \text{ and } |x - y| \geq 5.6 \end{cases} \quad (2)$$

$rating$ has been plotted in Fig. 1.

Consider an arbitrary frontal impact score x in the open interval $(2.2, 7.8)$ and a corresponding side impact score y such that $x - 5.6 < y < x + 5.6$. By definition, $rating(x, y) = 2$, but $rating(x, 16) = 1$. Since $y < 16$ follows from the above inequalities, the claim holds for the arbitrary pair $(x, y), (x, 16)$. Hence, proved.



*Latin NCAP, 2018-19

Figure 1: Stars vs. scores of vehicle models missing features required for 3 stars. Data collected from Global NCAP India and Africa projects, and 2016-19 Latin NCAP.

Observe that many cars would have had fewer stars if their side impact protection had been better. For example, the 2-star Maruti Suzuki Alto K10 would have shared a 1-star rating with the Suzuki Ertiga, had its side impact protection been better. Both have nearly identical frontal impact protection across body parts.

3 Appendix: Global NCAP limits

- Maximum score is 34.00 — front (16.00) + side (16.00) + seatbelt reminders (2.00) [1]
- Score thresholds for **1, 2, 3, 4 and 5** stars are **4.00, 10.00, 16.00, 22.00 and 27.00** points respectively. [1]
- A penalty of -1 star is applied if and only if

$$|x - y| \geq 5.6 \tag{3}$$

- Additional requirements (ESC, curtain airbags, etc.) are listed in [1].

References

- [1] Global NCAP. Adult Occupant Protection Assessment Protocol v2.0, 2021.